

Genetic Screens

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BINF5501 – Bioinformatics & Precision Medicine

Lecture 5



HUMBER

Faculty of Health Sciences & Wellness

Summary of Lecture 4

- The presence or absence of a genetic variant can dramatically impact a patient's response to drug treatment
- Chemotherapy targets essential cellular processes and preferentially targets dividing cells
- Molecularly targeted therapy offers a more rational approach to treating cancer but is rarely curative 
- Combination therapies can increase response rates in cancer patients by combatting patient-to-patient and intratumor heterogeneity
- Matching patients to biomarker-driven clinical trials can improve outcomes in a subset of patients 

Today's Lecture and Discussion Topics

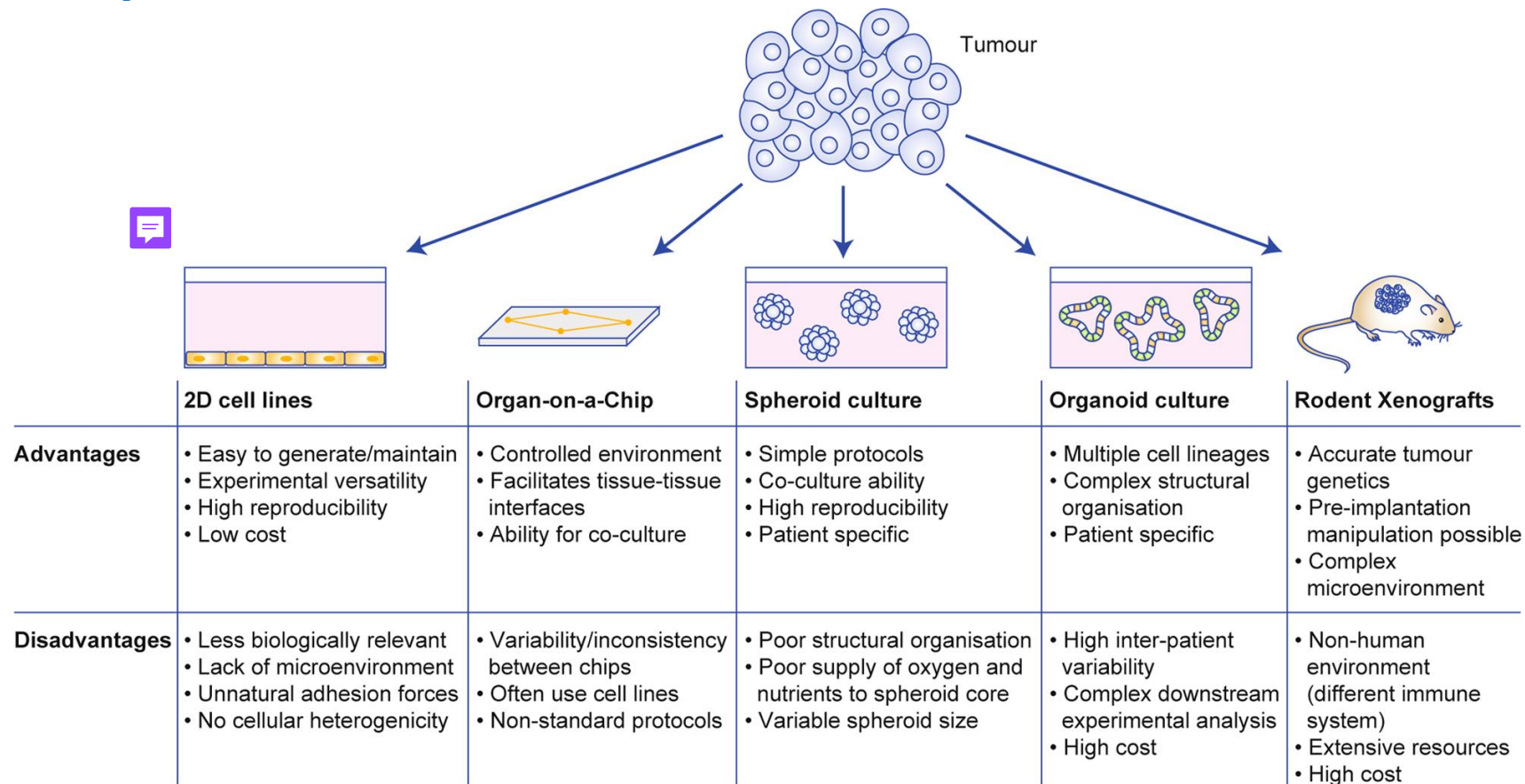
- Functional genomics
- Molecular and phenotypic assays
- Arrayed and pooled screens
- Off-target effects and the importance of controls

What is Functional Genomics?

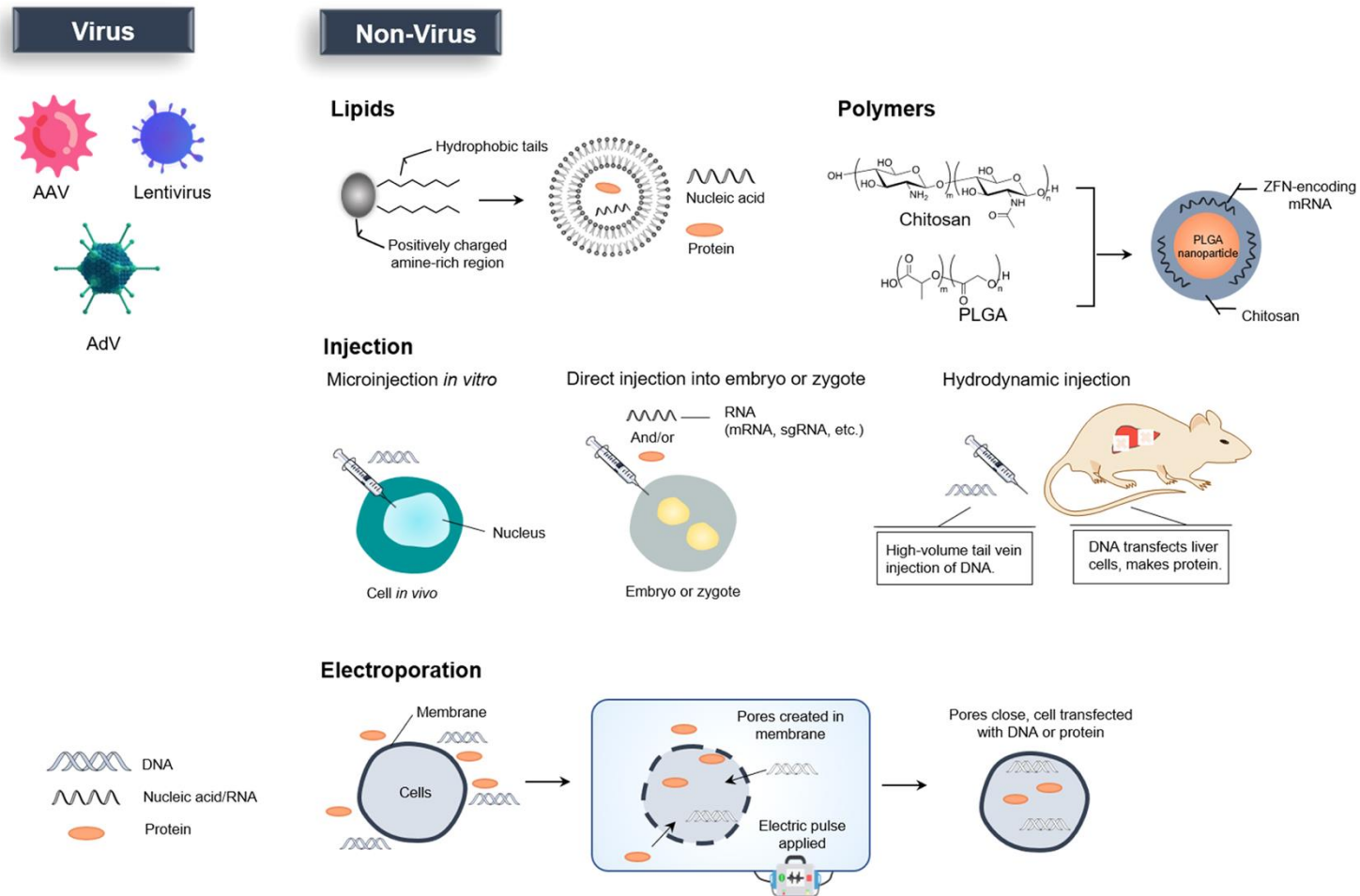
While traditional genomics considers the **sequence and structure** of genomes, functional genomics consider the **functions and interactions** of genes and genetic variants.

Functional genomic approaches frequently deploy experimental model systems and high-throughput screening to interrogate gene function.

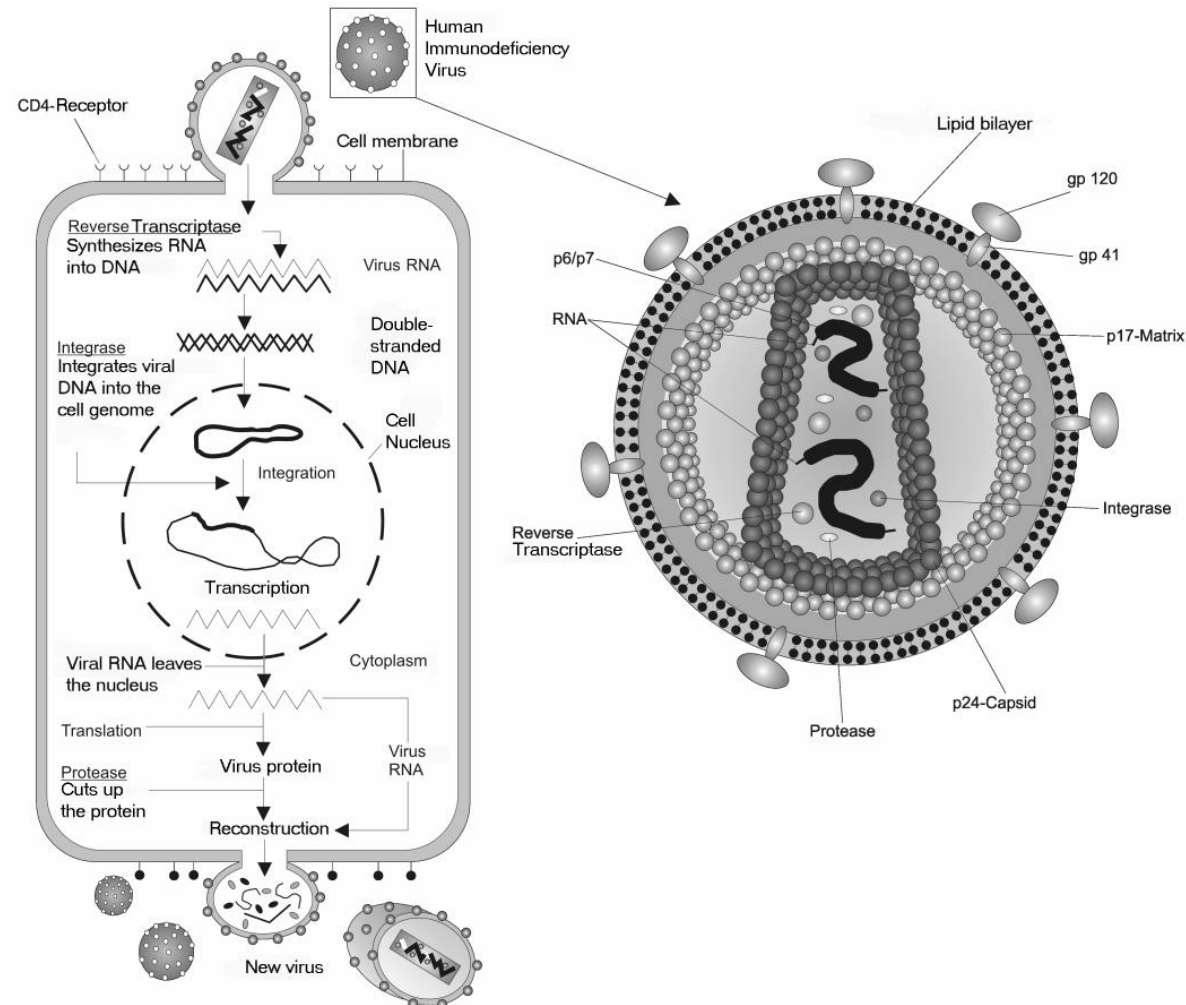
Experimental models of human cancer



Gene delivery systems for mammalian cells

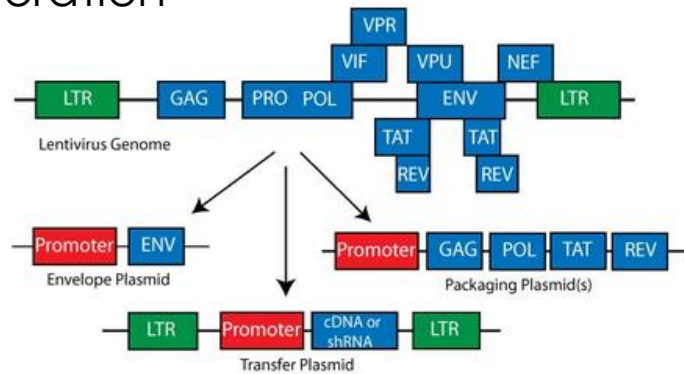


Retrovirus structure and life cycle

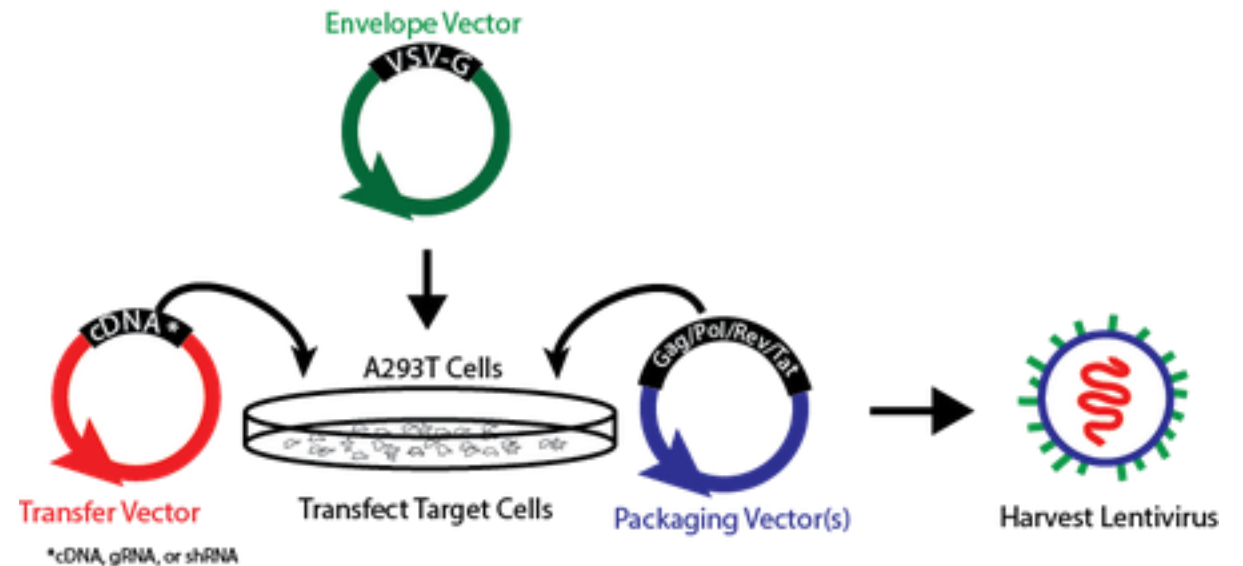
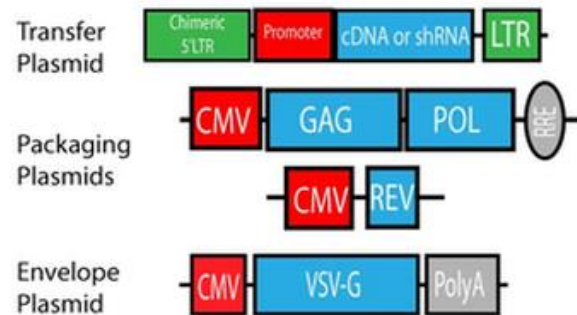


Lentiviruses are recombinant, non-replicating viruses derived from HIV-1

2nd generation



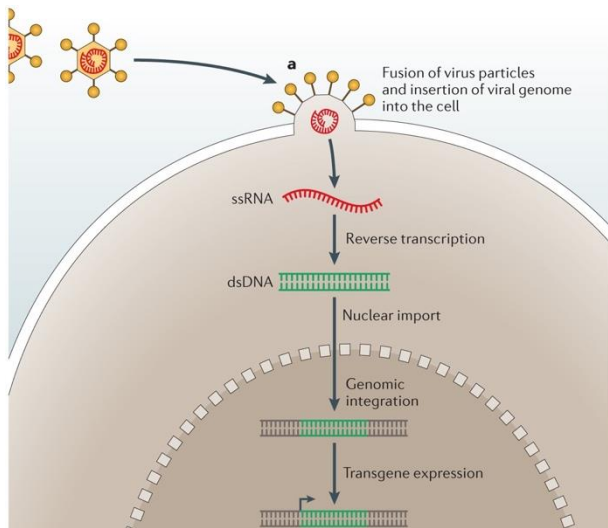
3rd generation



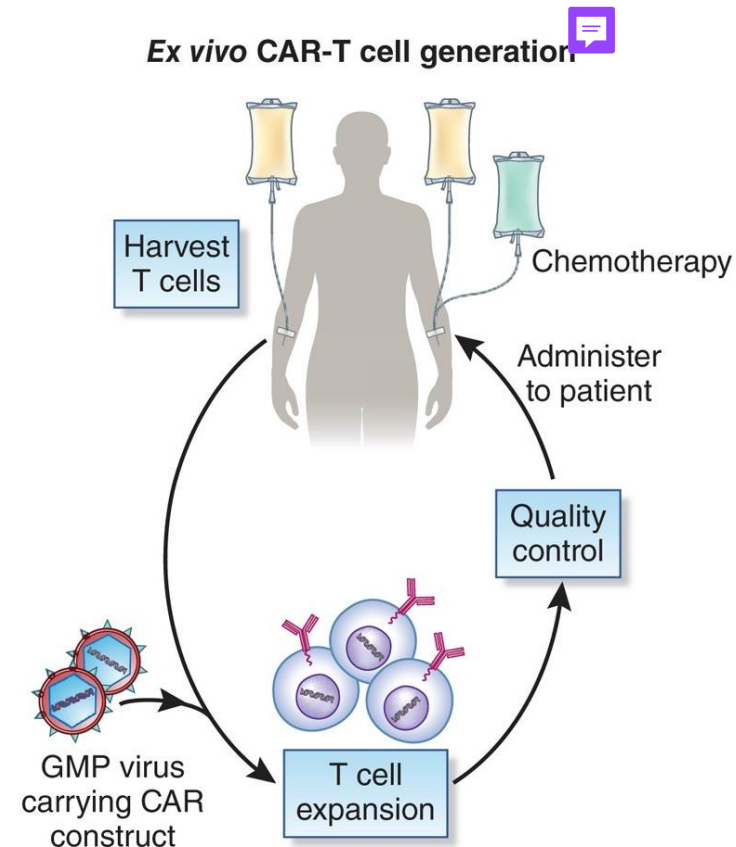
Genetic elements that enable lentiviral packaging and proviral integration

Plasmid Type	Element	Delivery relative to transgene	Purpose	https://www.addgene.org/guides/lentivirus/
Envelope	VSVG	<i>in trans</i>	Vesicular stomatitis virus G glycoprotein; Broad tropism envelope protein used to pseudotype most lentiviral vectors.	
Packaging	Gag	<i>in trans</i>	Precursor structural protein of the lentiviral particle containing Matrix, Capsid, and Nucleocapsid components.	
	Pol	<i>in trans</i>	Precursor protein containing Reverse Transcriptase and Integrase components.	
	Rev	<i>in trans</i>	On separate plasmid from Gag/Pol in third generation system; Binds to the Rev Response Element (RRE) within unspliced and partially spliced transcripts to facilitate nuclear export.	
	Tat	<i>in trans</i>	Second generation only; Trans-activator; binds TAR to activate transcription from the LTR promoter.	
Transfer	cPPT	<i>in cis</i>	Central polypurine tract; recognition site for proviral DNA synthesis. Increases transduction efficiency and transgene expression.	
	Psi (Ψ)	<i>in cis</i>	RNA target site for packaging by Nucleocapsid.	
	RRE	<i>in cis</i>	Rev Response Element; sequence to which the Rev protein binds.	
	WPRES	<i>in cis</i>	Woodchuck hepatitis virus post-transcriptional regulatory element; sequence that stimulates the expression of transgenes via increased nuclear export.	
	LTRSub-components: •U3 •RSub-element: • TAR •U5	<i>in cis</i>	LTR; Long terminal repeats; U3-R-U5 regions found on either side of a retroviral provirus (see below). Cloning capacity between the LTRs is ~8.5kb, but inserts bigger than ~3kb are packaged less efficiently. Sub-components: U3; Unique 3'; region at the 3' end of viral genomic RNA (but found at both the 5' and 3' ends of the provirus). Contains sequences necessary for activation of viral genomic RNA transcription. R; Repeat region found within both the 5' and 3' LTRs of retro/lentiviral vectors. Tat binds to this region. Sub-element: TAR; 2nd generation only; Trans-activating response element; located in the R region of the LTR and acts as a binding site for Tat. U5; Unique 5'; region at the 5' end of the viral genomic RNA (but found at both the 5' and 3' ends of the provirus).	
	5' LTR	<i>in cis</i>	Acts as an RNA pol II promoter. The transcript begins, by definition, at the beginning of R, is capped, and proceeds through U5 and the rest of the provirus. Third generation vectors use a hybrid 5' LTR with a constitutive promoter such as CMV or RSV.	
	3' LTR	<i>in cis</i>	Terminates transcription started by 5' LTR by the addition of a poly A tract just after the R sequence.	

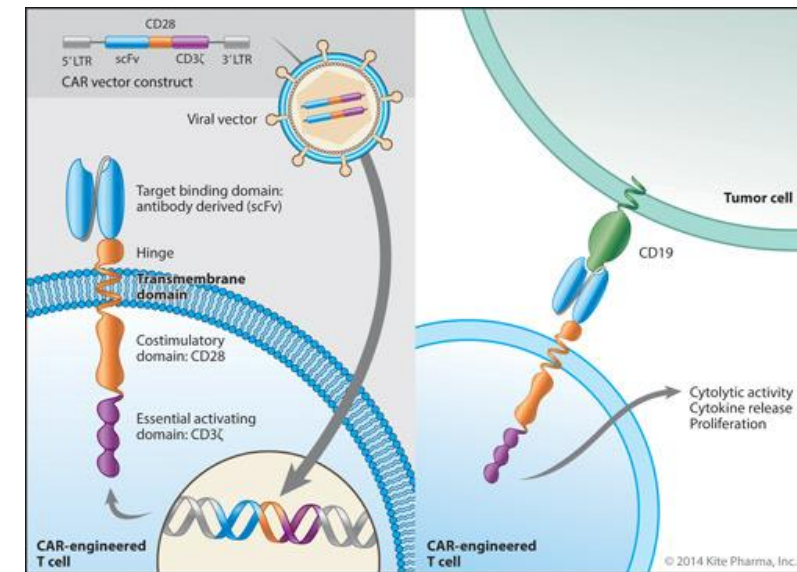
Lentiviral vectors enable efficient engineering of human cells *ex vivo*



Shalem et al. Nature Reviews Genetics volume 16, pages 299–311 (2015)
<https://www.nature.com/articles/nrg3899>

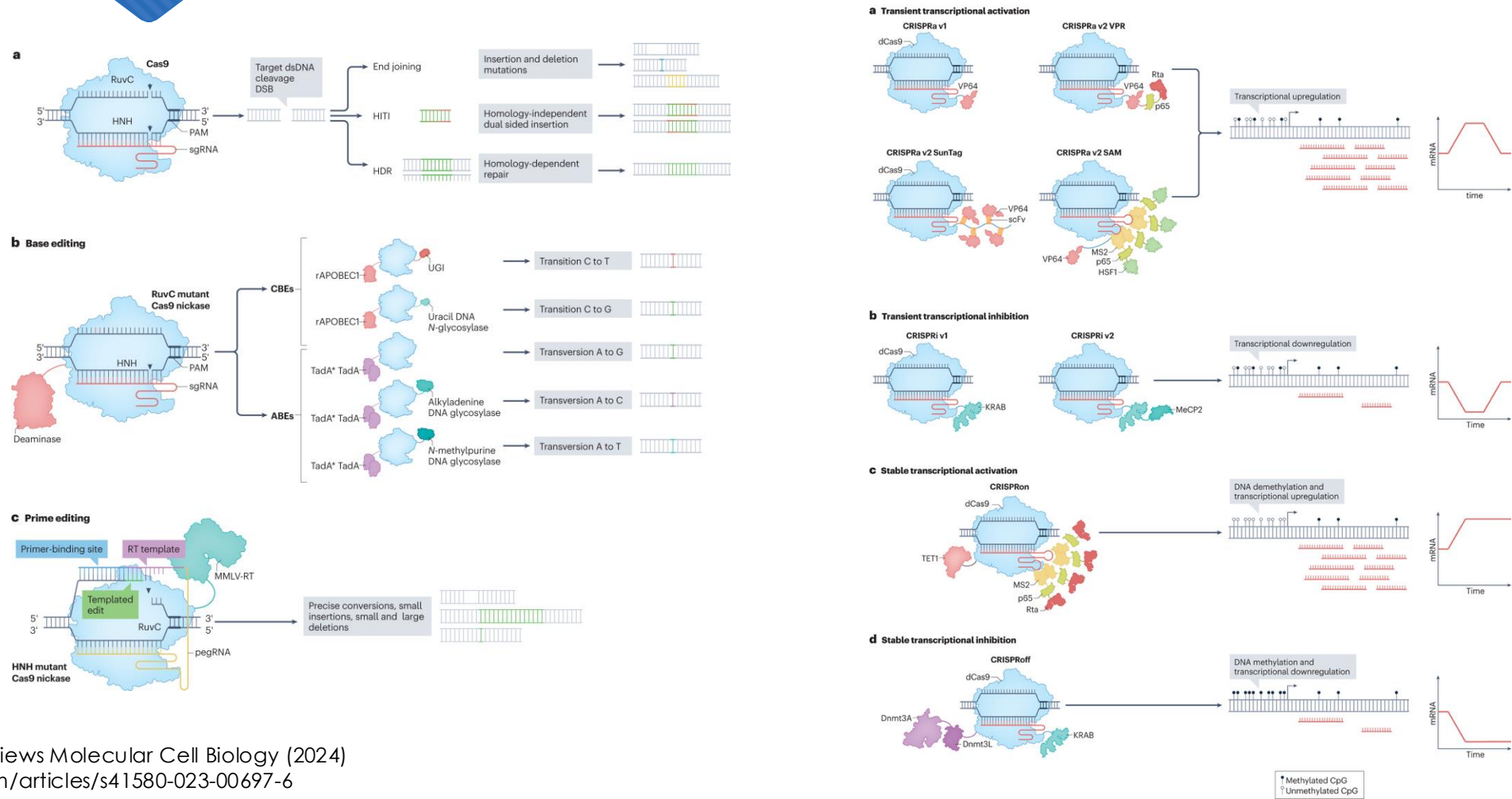


Olweus. Nature Biotechnology volume 35, pages 520–521 (2017)
<https://www.nature.com/articles/nbt.3898>



<https://www.genengnews.com/insights/know-your-enemy-takes-cell-therapy-to-next-level/>

Genome and epigenome engineering using CRISPR/Cas9-based technologies



Discussion Question

What are genetic screens and what are they good for?



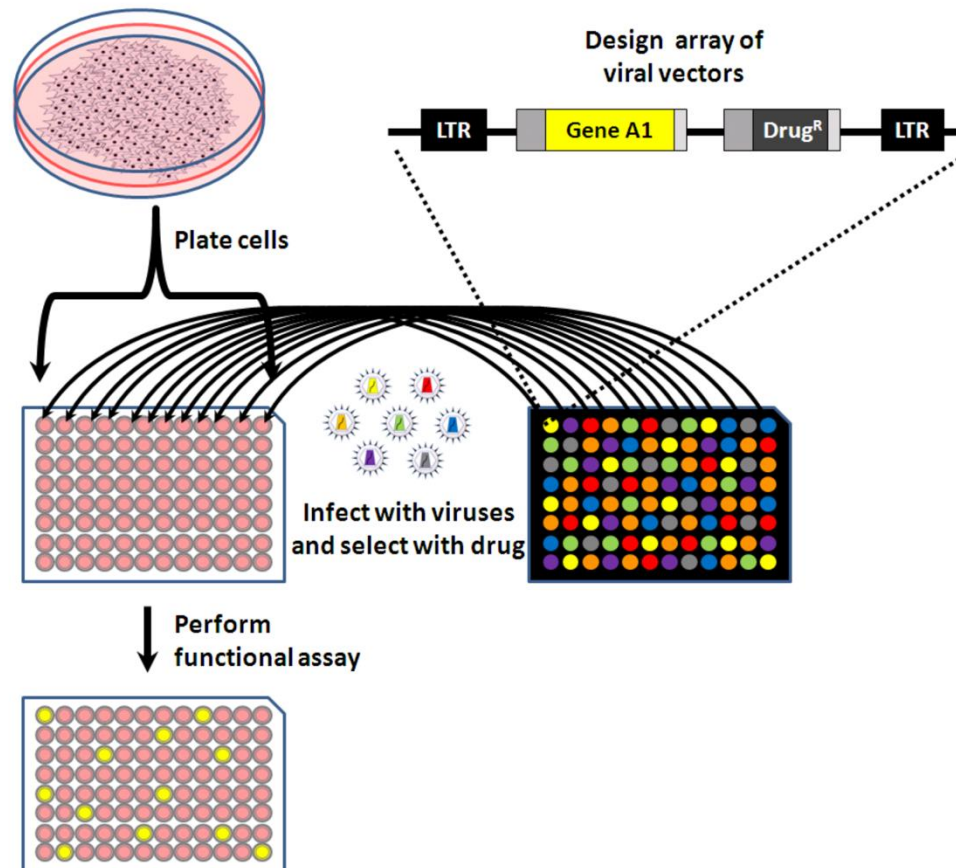
Molecular and phenotypic assays

- A **molecular assay** measures a change in molecular composition
- Often focused on a particular protein, signaling pathway, or transcriptional output
- Does not strictly provide information on disease phenotype
- Examples:
 - RNA/Protein marker expression
 - Transcriptional reporter assays
 - Single-cell RNA-Seq & Multiomics
 - **Cell Painting**
- A **phenotypic assay** measures a change in cellular or organismal phenotype
- Often disease-focused
- Does not strictly provide information on molecular mechanism
- Examples:
 - Cell viability/proliferation/motility
 - Size/shape/color
 - Animal behavior
 - Disease onset/symptoms/severity

Loss-of-Function (LOF) and Gain-of-Function (GOF) screens

- In **loss-of-function** screens, cells or organisms are engineered to **suppress** or **inactivate** an intact gene or genetic element.
- Examples include:
 - shRNA
 - siRNA
 - CRISPR
 - CRISPRi
 - Direct Mutagenesis
- In **gain-of-function** screens, cells or organisms are engineered to **overexpress** or **hyperactivate** a gene or genetic element.
- Examples include:
 - cDNA
 - CRISPRa
 - Direct mutagenesis

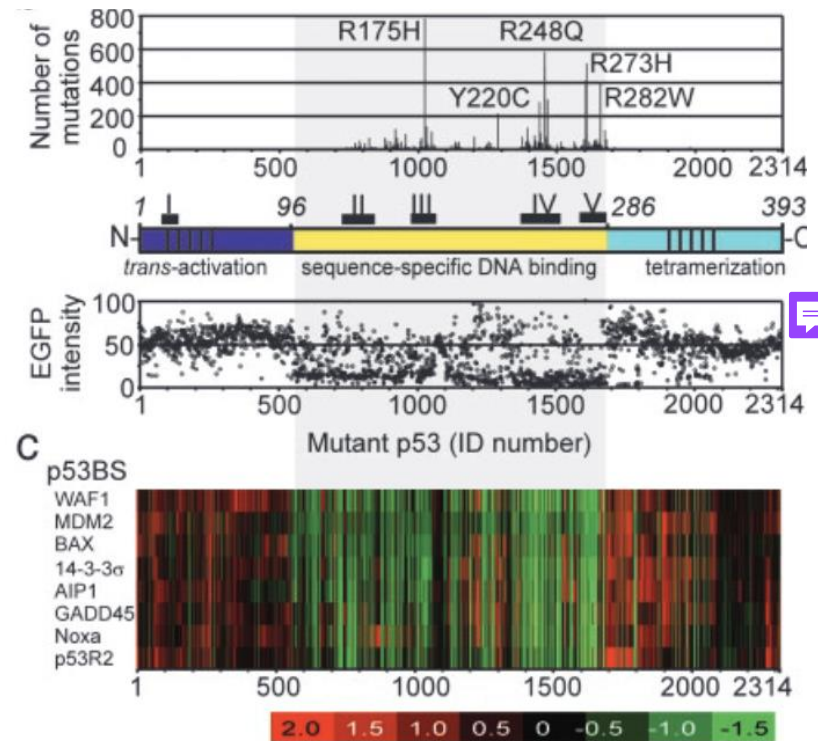
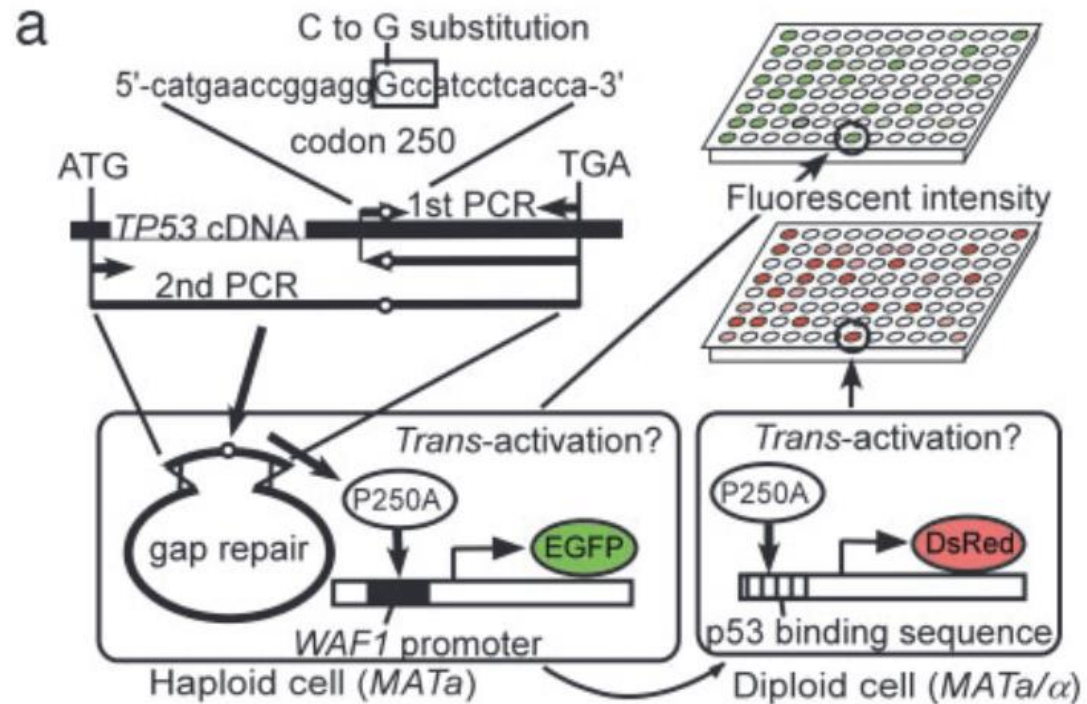
Arrayed screens enable individual perturbations to be tested in isolation



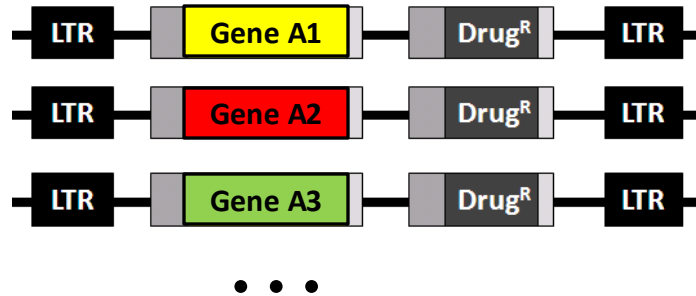
Any vector

- Constitutive or inducible promoter driving cDNA, shRNA, or sgRNA expression
- Drug resistance cassette is optional 💬
- Can infect at a low or high multiplicity of infection (MOI)

An arrayed molecular screen to identify p53 mutations that affect transactivation potential

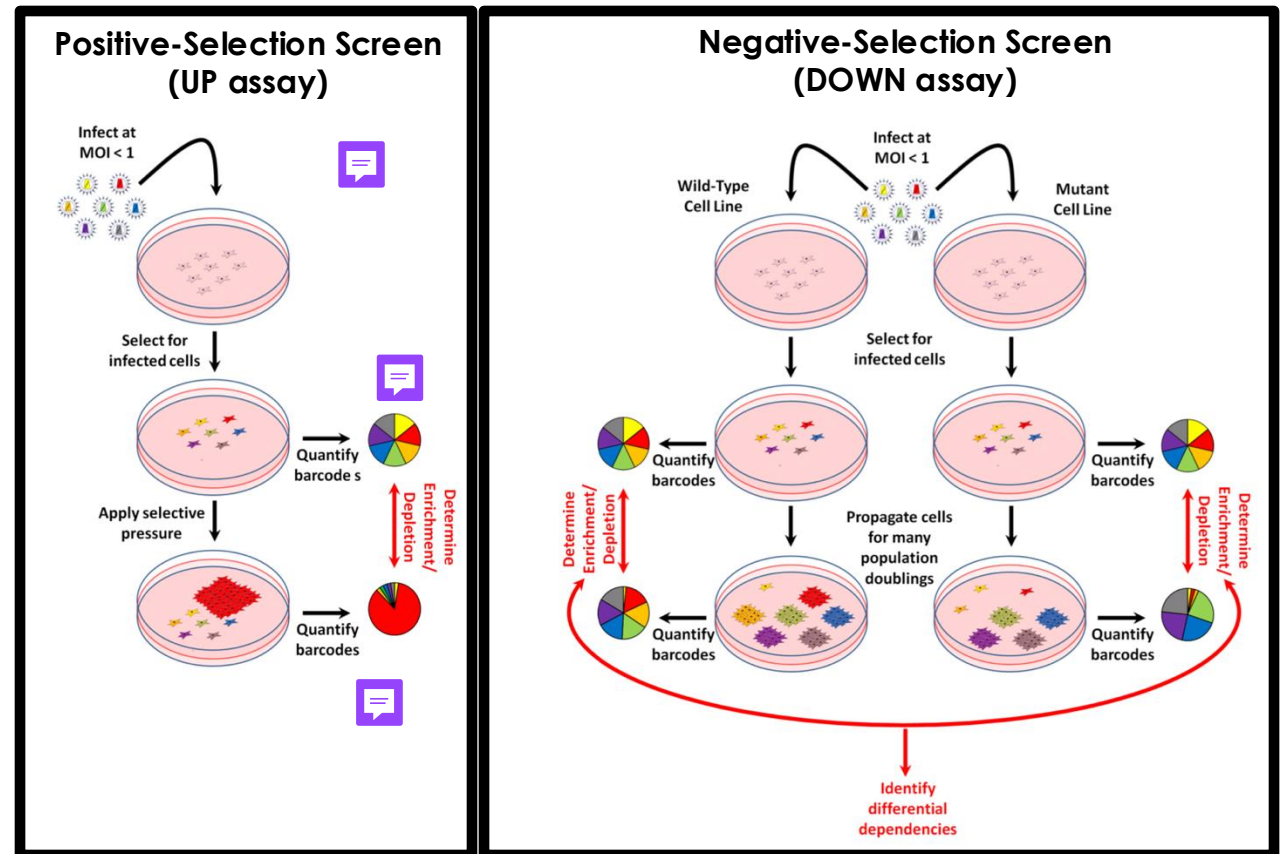


Pooled screens enable thousands of perturbations to be tested in parallel



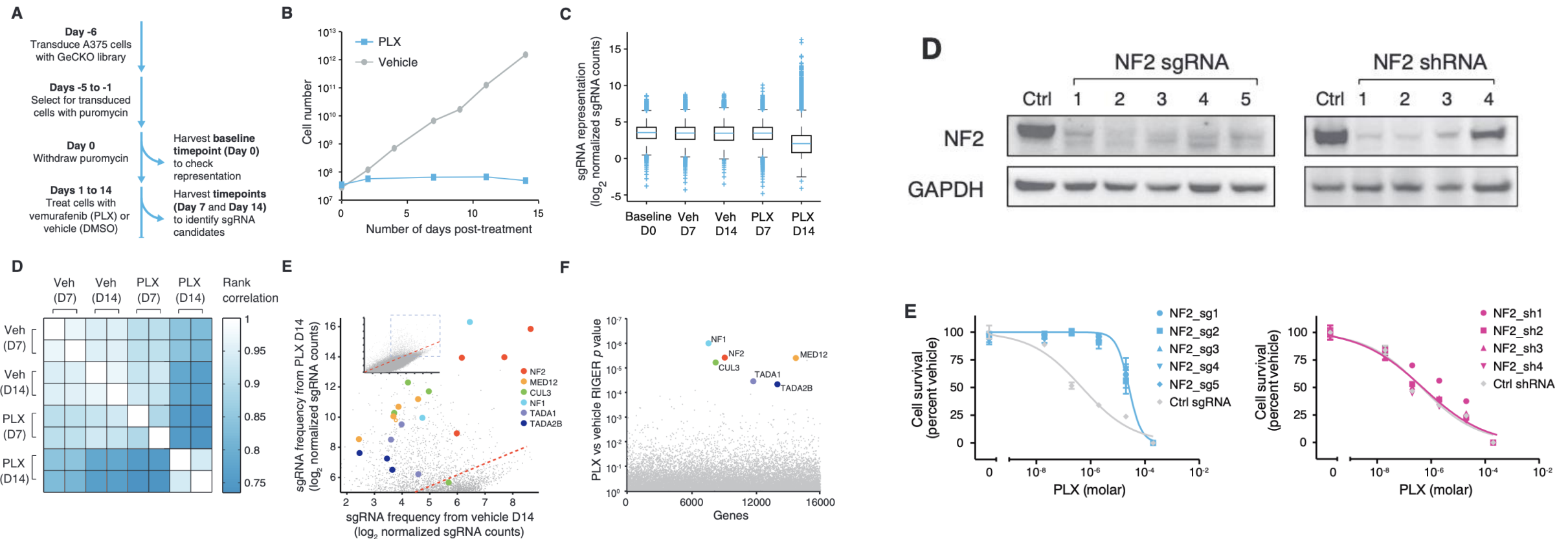
Integrating lentiviral vector

- Constitutive promoter driving cDNA, shRNA, or sgRNA expression
- Drug resistance cassette
- Aim for at most 1 integration event per cell (multiplicity of infection < 1)



LOF vs. GOF is independent of negative-selection vs. positive-selection

A pooled positive selection phenotypic screen to identify genes that confer resistance to vemurafenib



Summary

- Genetic screens enable unbiased discovery of genes or variants that contribute to a molecular or phenotypic change.
- Genetic perturbations can be screened individually (arrayed) or in parallel (pooled).
- Generally, loss-of-function screens and negative selection screens need to be scrutinized more intensely than gain-of-function screens and positive selection screens.



Questions?